

EDUCATION

Vanderbilt University

Ph.D. in Biostatistics, GPA: 3.86/4.00

Nashville, TN

Expected May 2029

Wake Forest University

M.S. in Statistics, GPA: 4.00/4.00

Graduate Certificate in Data Science, GPA: 4.00/4.00

Thesis Advisor: Sarah Lotspeich, Ph.D.

- Thesis: Adjusting for covariate misclassification to quantify the relationship between diabetes and local access to healthy food

Winston-Salem, NC

May 2024

University of Scranton

B.S. in Applied Mathematics and B.A. in Philosophy, *Summa Cum Laude*, GPA: 3.94/4.00

Minors in Computer Science and Finance, Concentration in Data Science

Honors Program, Special Jesuit Liberal Arts Honors Program

Thesis Advisor: Murong Xu, Ph.D.

- Thesis: Music Feature Classification: Using Techniques of Machine Learning to Detect Genre
- Available from <https://digitalservices.scranton.edu/digital/collection/p15111coll1/id/1340/rec/68>

Scranton, PA

May 2022

St. Anthony's High School

Advanced Regents Diploma

South Huntington, NY

June 2018

RESEARCH AND WORK EXPERIENCE

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| 10. Research Assistant
<i>Vanderbilt University Medical Center, Department of Biostatistics</i>
Mentor: Rameela Raman, Ph.D. | Nashville, TN
July 2024 – Present |
| 9. Lead Teaching Assistant
<i>Wake Forest University, Departments of Statistical Sciences and Mathematics</i>
Mentor: Lynne Yengulalp, Ph.D. | Winston-Salem, NC
Mar 2023 – May 2024 |
| 8. Research Assistant
<i>Wake Forest University, Department of Statistical Sciences</i>
Mentor: Sarah Lotspeich, Ph.D. | Winston-Salem, NC
Jan 2023 – May 2024 |
| 7. Data Science Intern
<i>Hanesbrands</i> | Winston-Salem, NC
June 2023 – August 2023 |
| 6. Academic Tutor
<i>Self-Employed</i> | Huntington, NY
January 2012 – Present |
| 5. Summer Research Intern
<i>Central Washington University, Department of Mathematics</i>
Mentor: Sooie-Hoe Loke, Ph.D. | Virtual
June 2021 – August 2021 |

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| 4. Summer Research Intern
<i>Carnegie Mellon University, Department of Statistics and Data Science</i>
Mentors: Sam Ventura, Ph.D. and Ron Yurko, Ph.D. | Virtual
June 2020 – August 2020 |
| 3. Academic Tutor
<i>University of Scranton, Center for Teaching and Learning Excellence</i> | Scranton, PA
March 2019 – May 2022 |
| 2. Financial Analyst
<i>VentureBrandUSA LLC</i> | East Northport, NY
July 2019 – July 2021 |
| 1. Research Intern
<i>Long Island University, Department of Biology</i>
Mentors: Scott Santagata, Ph.D. and Paul Paino | Brookville, NY
March 2017 – August 2017 |

HONORS AND AWARDS

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| 18. Graduate Leadership Anchor Award, Vanderbilt University
Awarded annually for outstanding mentorship | 2026 |
| 17. Women in Statistics and Data Science Travel Award
Awarded by the ASA Biopharmaceutical Section for outstanding achievements and leadership in the field | 2025 |
| 16. SPAIG Award, American Statistical Association
Given annually to recognize outstanding collaborations among academe, industry, and government organizations while recognizing key individual contributors | 2025 |
| 15. Highly Commended Article, Significance Magazine
Designation from the annual Statistical Excellence Award for Early-Career Writing Competition | 2024 |
| 14. Russell G. Hamilton Scholar, Vanderbilt University | 2024-2029 |
| 13. Provost's Graduate Fellowship, Vanderbilt University | 2024-2029 |
| 12. Graduate Student Award for Outstanding Scholarship, Wake Forest University
Awarded by the Department of Statistical Sciences for scholarship, ingenuity, and dedication | 2024 |
| 11. Travel Award, Women in Statistics and Data Science Conference
Merit-based award for student or early professional conference attendees | 2023, 2024, 2025 |
| 10. Gertrude M Cox Scholarship, American Statistical Association
Awarded for dedication to statistical inquiry, passion for mentoring, and extraordinary commitment to scholarly excellence | 2023 |
| 9. Teaching Assistantship, Wake Forest University
Full tuition grant and teaching stipend | 2022 – 2024 |
| 8. Provost Merit Scholarship, Wake Forest University
Merit-based graduate student award | 2022 – 2024 |
| 7. Outstanding Poster, Mathematical Association of America MathFest
Awarded for excellent project presentation and poster design | 2022 |
| 6. Excellence in Applied Mathematics Award, University of Scranton
Merit-based department award | 2022 |
| 5. Humanities Fellow, University of Scranton Slattery Center
Undergraduate humanities research grant | 2021 – 2022 |
| 4. Presidential Scholarship, University of Scranton
Merit-based full-tuition scholarship awarded for exemplary academic achievement | 2018 – 2022 |

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| 3. Dean's List, University of Scranton
Distinction granted to full-time students with a GPA above 3.50 | 2018 – 2022 |
| 2. Central Convergence REU Scholar
Undergraduate research grant | 2021 |
| 1. National Hispanic Scholar, Hispanic Scholarship Fund
Merit-based distinction and scholarship awarded to university students of Hispanic heritage | 2018 – 2020 |

PEER-REVIEWED PUBLICATIONS

2. **A.E. Mullan**, P.D.A. Nguyen, S.C. Lotspeich, “Linking potentially misclassified healthy food access to diabetes prevalence,” *Statistics and Public Policy*, in press, 2026+. doi:10.1080/2330443X.2026.2667842
1. S.C. Lotspeich, **A.E. Mullan**, L. D’Agostino McGowan, and S.A. Hepler, “Combining straight-line and map-based distances to investigate the connection between proximity to healthy foods and disease,” *Statistics in Medicine*, vol. 44, no. 7, pp. e70054, 2025. doi:10.1002/sim.70054.

PROFESSIONAL SERVICE PUBLICATIONS

1. **A.E. Mullan**, “For better or for worse: The “kiss effect” on television ratings,” *Significance*, vol. 22, no. 6, pp. 30-33, 2025. <https://doi.org/10.1093/jrssig/qmaf075>

MEDIA

1. Feb. 2025. Interview. For better or for worse: The “kiss effect” on television ratings. Stats + Stories Podcast.

PRESENTATIONS

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| 17. Contributed Talk: Linking Potentially Misclassified Healthy Food Access to Diabetes Prevalence - <i>ENAR Spring Meeting</i> | Indianapolis, IN
March 2026 |
| 16. Invited Talk: Linking Potentially Misclassified Healthy Food Access to Diabetes Prevalence - <i>Women in Statistics and Data Science Conference</i> | Cincinnati, OH
November 2025 |
| 15. Invited Talk: Talking Over a Football Game: The Taylor Swift Effect on Football Podcast Popularity - <i>International Day of Women in Statistics and Data Science</i> | Virtual
October 2025 |
| 14. Invited Talk: Modeling Misclassification: Exploring the Relationship between Diabetes and Access to Healthy Foods - <i>International Day of Women in Statistics and Data Science</i> | Virtual
October 2025 |
| 13. Contributed Talk: A Two-Pronged Approach for Identifying and Addressing Problematic Psychotropic Prescribing in the Child Welfare System (joint presentation with T. Kuhn)
<i>Transformational Collaborative Outcomes Management Conference</i> | Chicago, IL
October 2025 |
| 12. Contributed Poster and Speed Talk: Linking Potentially Misclassified Healthy Food Access to Diabetes Prevalence - <i>Joint Statistical Meetings</i> | Nashville, TN
August 2025 |
| 11. Contributed Poster: Visualizing Cost Effectiveness Analysis with Second-Generation Acceptability Curves - <i>ENAR Spring Meeting</i> | New Orleans, LA
March 2025 |
| 10. Invited Talk: Misclassification Challenges in Quantifying the Relationship Between Diabetes and Local Access to Healthy Food
<i>Department of Mathematics, University of Scranton</i> | Scranton, PA
March 2025 |

9. **Contributed Poster:** Adjusting for covariate misclassification to quantify the relationship between diabetes and local access to healthy food
Women in Statistics and Data Science Conference Reston, VA
October 2024
8. **Invited Talk:** For Better or For Worse: The First Kiss Effect on Television Ratings
International Day of Women in Statistics and Data Science Virtual
October 2024
7. **Contributed Talk:** Adjusting for Measurement Error to Quantify the Relationship Between Diabetes and Local Access to Healthy Food - *ENAR Spring Meeting* Baltimore, MD
March 2024
6. **Contributed Poster:** Rate This Interruption: Using Interrupted Time Series Techniques to Analyze Popular Television Couples and Episode Ratings
Women in Statistics and Data Science Conference Bellevue, WA
October 2023
5. **Contributed Poster:** Music Feature Classification: Using Techniques of Machine Learning to Detect Genre - *Mathematical Association of America MathFest Conference* Philadelphia, PA
August 2022
4. **Contributed Poster:** Deal or No Deal: Modeling a Game Show with Utility Theory and Machine Learning - *Joint Mathematics Meetings* Virtual
April 2022
3. **Contributed Poster:** Deal or No Deal: Modeling a Game Show with Utility Theory and Machine Learning - *SACNAS NDiSTEM Conference* Virtual
August 2021
2. **Featured Talk:** Deal or No Deal: Modeling a Game Show with Utility Theory and Machine Learning - *Mount Stuart Mathematics Seminar Series* Virtual
August 2021
1. **Featured Talk:** A Puck Above the Rest: The Effects of New Data on 2020 NHL Draft Decisions
Carnegie Mellon Sports Analytics Conference Virtual
October 2020

GUEST LECTURES AND TUTORIALS

7. From Idea to Conclusion: A Statistical Perspective
Guest lecturer for the School for Science and Mathematics at Vanderbilt, Vanderbilt University Nashville, TN
March 2026
6. Ask not what you can do for math, but what math can do for you!
Guest speaker for the Math Honor Society, Saint Anthony's High School Virtual
February 2026
5. The Statistician's Storybook: Crafting Accessible and Engaging Presentations
Teaching assistant for tutorial at the Women in Statistics and Data Science Conference Virtual
October 2025
4. Statistics in Your Own Backyard
Guest lecturer for Day of Discovery program, Vanderbilt University Nashville, TN
September 2025
3. Statistical Ideas You Can Never Unsee
Guest lecturer for PSYCH1130, Cornell University Ithaca, NY
March 2025
2. From Idea to Conclusion: A Statistical Perspective
Guest lecturer for the School for Science and Mathematics at Vanderbilt, Vanderbilt University Nashville, TN
February 2025
1. Stats can be fun, I swear!
Guest lecturer for PSYCH1130, Cornell University Virtual
September 2024

PANEL DISCUSSIONS

5. **Panelist:** Careers in Statistics and Data Science
Saint Anthony's High School Virtual
March 2026
4. **Moderator:** Beyond Academia - Exploring Statistics Careers in Industry, Government, and More
Joint Statistical Meetings Nashville, TN
August 2025

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| 3. Panelist: Graduate Admissions Advice
<i>Electronic Undergraduate Student Research Conference</i> | Virtual
November 2024 |
| 2. Moderator: Statistical Storytelling: Insights Into Effective Presentation Strategies
<i>Women in Statistics and Data Science Conference</i> | Reston, VA
October 2024 |
| 1. Moderator: The Art of the Invite: Crafting Successful Invited Session Proposals
<i>International Day of Women in Statistics and Data Science Conference</i> | Virtual
October 2024 |

SOFTWARE

5. **Author:** wickedplyr - R Package
<https://github.com/ashleymullan/wickedplyr>
4. **Author:** Stats can be fun, I swear! - Shiny App
<https://ashley-mullan.shinyapps.io/lecture-app>
3. **Author:** visegen - R Package
<https://github.com/ashleymullan/visegen>
2. **Contributor:** POSSUM - R Package
<https://github.com/sarahlotspeich/possum>
1. **Contributor:** ThemePark - R Package
<https://github.com/MatthewBJane/ThemePark>

TEACHING

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| 4. Vanderbilt Summer Academy
Instructor: Seize the Data - Intro to Statistical Thinking | Nashville, TN
July 2026 |
| 3. Wake Forest University
Teaching Assistant: STA111 - Elementary Probability and Statistics
Supervising Professor: Leo Cella, Ph.D. | Winston-Salem, NC
Aug 2022 – May 2023 |
| 2. University of Scranton
Teaching Assistant: INTD 290 - Leadership and Civic Responsibility
Supervising Professors: Teresa Grettano, Ph.D. and Stephen Whittaker, Ph.D. | Scranton, PA
May 2020 – Aug 2020 |
| 1. University of Scranton
Sectional Assistant: Performance Music
Supervising Instructor: Cheryl Boga | Scranton, PA
Aug 2018 – May 2022 |

MENTORSHIP

Research Supervision

High School Students

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| — Danny Hoberman | Jun 2025–November 2025 |
| * Selected as 2026 Regeneron STS Top 300 Scholar | |
| * Invited Talk at International Day of Women in Statistics and Data Science 2025 | |
| * <u>Next Position:</u> B.S. Student at Duke University | |
| — Carrie LaBarca | Jul 2025–November 2025 |
| * Invited Talk at International Day of Women in Statistics and Data Science 2025 | |

PROFESSIONAL SERVICE AND OUTREACH

Officer Positions and Committees

Abbreviations: Eastern North American Region of the International Biometrics Society (ENAR), Biostatistics Graduate Student Association (BGSA), International Conference on Health Policy Statistics (ICHPS), Joint Statistical Meetings (JSM)

- Founding Member, Undergraduate Task Force, ENAR 2026–2027
- Member, Outreach Committee, ICHPS 2026–2027
- Member, JSM Docent Committee 2026–
- Co-Organizer, JSM Data Art Show 2024–
- BGSA, Vanderbilt University 2025–2027
 - * President 2026–2027
 - * Communications Officer/First Year Liaison 2025–2026
- Founding Member, Mentorship Committee, ENAR 2025–
- Member, Program Committee, Caucus for Women in Statistics 2025–
- Council for Emerging and New Statisticians, ENAR 2024–2027
 - * Co-Chair 2026–2027
 - * Steering Committee, Social Media Co-Chair 2025–2026
 - * Social Media Co-Chair 2024–2025
- Social Media Committee, ENAR 2023–2026
 - * Chair 2025–2026
 - * Member 2023–2025
- Member, Graduate Honor Council, Wake Forest University 2022–2024
- President, Pi Mu Epsilon Honor Society, University of Scranton 2021–2022
- Secretary, Phi Sigma Tau Honor Society, University of Scranton 2021–2022
- Secretary, Upsilon Pi Epsilon Honor Society, University of Scranton 2021–2022
- President, Mathematics Club, University of Scranton 2018–2022
- Vice President, HackScranton University of Scranton 2019–2021

Volunteer Outreach

- Judge, ASA K-12 Data Visualization Poster Competition 2025
- Judge, Undergraduate Statistics Project Competition 2025, 2026
- Mentor, Flash Mentoring, Caucus for Women in Statistics 2024, 2025, 2026
- Pen Pal, Letters to a Pre-Scientist 2024–
- Activity Lead, Florence Nightingale Day, Wake Forest University 2024
- Proctor, MathCounts, Wake Forest University 2023
- Dean’s Ambassador, College of Arts and Sciences, University of Scranton 2021–2022
- Admission Ambassador, University of Scranton 2019–2022
- Royal Tie Peer Mentor, University of Scranton 2021–2022

Conference Activities

- Invited Session Organizer (Frequency)
 - * *ENAR Spring Meeting* (2), *Women in Statistics and Data Science* (3), *International Day of Women in Statistics and Data Science* (4)
- Session Chair (Frequency)
 - * *Joint Statistical Meetings* (1), *International Day of Women in Statistics and Data Science* (3), *ENAR Spring Meeting* (2), *Women in Statistics and Data Science* (3), *Caucus for Women in Statistics Webinar* (1)

SKILLS

- **Programming:** R (advanced), Quarto (advanced), SQL (intermediate), Python (intermediate), LaTeX (intermediate), GitHub (intermediate), Shiny (intermediate), Java (intermediate), Cluster Computing (novice), Julia (novice), Power BI (novice), HTML (novice)
- **Applications:** Microsoft Suite (advanced), Google Suite (advanced), Bloomberg Terminal (novice)
- **Languages:** English (native), Spanish (intermediate written and oral proficiency)